

# **Fingerprint Image Enhancement and Quality Assessment System**

## **BMDS2133 Image Processing Assignment Documentation**

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**Group Members :**

- |                    |                                        |
|--------------------|----------------------------------------|
| (a) Mok Chun Bing  | Non-Local Means (NLM) Denoising        |
| (b) Lee Mun Hinn   | Modified Gabor Ridge Enhancement       |
| (c) Lam Zhan Foong | Total Variation (TV) Restoration       |
| (d) Lim Kai Xi     | Coherence-Guided Directional Diffusion |

**Programme :**

**Bachelor in Software Engineering (Honours) ▾**

**Tutor : Ms. Nurul Akmar binti Azman ▾**

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# Introduction

## 1.1 Problem Background

A fingerprint consists of distinctive patterns of ridges and valleys on the epidermal surface of a human fingertip. These patterns are formed before birth and remain largely permanent throughout an individual's lifetime, making fingerprints a widely adopted biometric characteristic for personal identification. Fingerprint recognition has consequently been applied in areas such as law enforcement, border control and consumer-device authentication (Maio & Maltoni, 1997).

Fingerprint image enhancement refers to digital image-processing techniques used to improve the visibility, clarity and continuity of fingerprint ridge-valley structures. Clear ridge structures are important because subsequent fingerprint-processing stages depend on the quality of the captured image. However, fingerprint images acquired using sensors or other acquisition methods may suffer from degradation caused by noise, low contrast, dry or wet skin, smudging and incomplete capture. Such degradation can obscure ridge patterns and reduce the reliability of fingerprint recognition systems (Hong, Wan, & Jain, 1998).

Fingerprint enhancement performance can be assessed using measurable image-quality characteristics. Quantitative metrics such as Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), and Mean Squared Error (MSE) can be used to determine how closely an enhanced image resembles its clean reference image (Gonzalez & Woods, 2017; Wang et al., 2004). In general, a better enhancement result is indicated by higher PSNR and SSIM values and a lower MSE value.

Various classical image-processing techniques can be applied to improve degraded fingerprint images. However, different enhancement algorithms respond differently to image degradation. A technique that effectively suppresses noise may not necessarily preserve ridge structures as effectively as another technique. Therefore, determining the most suitable enhancement approach requires controlled quantitative comparison using consistent data and evaluation metrics.

This project investigates four classical fingerprint enhancement techniques: **Non-Local Means (NLM) denoising, Modified Gabor Ridge Enhancement, Total Variation (TV) Restoration, and Coherence-Guided Directional Diffusion**. Their performance is evaluated using the publicly available **Sokoto Coventry Fingerprint Dataset (SOCOFing)** under a controlled comparative experiment. The techniques are evaluated using quantitative measures including Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Mean Squared Error (MSE), and processing runtime to determine their relative effectiveness for fingerprint image enhancement.

## 1.2 Objectives/Aims

- Objective 1:** To implement and evaluate four classical fingerprint enhancement techniques—Non-Local Means (NLM) Denoising, Modified Gabor Ridge Enhancement, Total Variation (TV) Restoration, and Coherence-Guided Directional Diffusion—using a 500-image Development Set from 50 SOCOFing subjects, evaluated using PSNR, SSIM, MSE, and processing runtime.
- Objective 2:** To select and finalise the configuration for each enhancement technique based on the Development Set results, and subsequently evaluate the selected configurations on 1,000 fingerprint images from 100 unseen SOCOFing subjects, with no subject or image overlap between the development and validation sets.
- Objective 3:** To deliver a practical Streamlit prototype that applies the selected enhancement techniques to uploaded fingerprint images, supports single-image, Compare All, and bulk workflows, and reports reference-based metrics only when a valid reference image is supplied.

### 1.3 Motivation

Reliable fingerprint image enhancement is important for supporting accurate and accessible biometric identification systems. As fingerprint-based authentication is increasingly used in digital services, mobile devices and identity verification systems, the quality of fingerprint images can directly influence the reliability of subsequent biometric processing. Classical image-processing techniques provide a practical approach to fingerprint enhancement because they can improve degraded images without requiring model training or large labelled datasets typically needed by deep-learning approaches.

The use of computationally efficient fingerprint enhancement techniques also supports the development of resilient and accessible digital infrastructure. The four techniques investigated in this project are based on classical image processing and can be implemented without the training overhead required by learning-based approaches. This makes them suitable for further investigation in resource-constrained or edge-computing environments. Consequently, the project aligns with Sustainable Development Goal 9: Industry, Innovation and Infrastructure, which promotes resilient infrastructure and technological innovation (United Nations, 2015), as illustrated in Figure 1.1.

Within the Malaysian context, reliable digital identification technologies can also contribute to the development of trustworthy digital public services. This is relevant to the **Malaysia MADANI** framework, particularly the core values of *Kemampanan* (Sustainability) and *Daya Cipta* (Innovation), which emphasise sustainable development and the empowerment of human creativity and innovation (Malaysia MADANI, n.d.), as illustrated in Figure 1.2. Improving the reliability of fingerprint image processing can contribute to biometric systems that support secure and efficient digital-service delivery.

From a commercialisation perspective, the fingerprint enhancement and quality assessment system developed in this project could serve as a lightweight preprocessing component for automated fingerprint identification systems (AFIS), biometric verification applications, or mobile authentication software development kits. The comparative evaluation of NLM, Modified Gabor, TV Restoration, and Coherence-Guided Directional Diffusion also provides

quantitative evidence for selecting an appropriate enhancement technique according to enhancement quality, structural preservation, and computational requirements.



*Figure 1.1: SDG9 Industry, Innovation and Infrastructure (United Nations, 2015)*



*Figure 1.2: Malaysia MADANI Logo (Malaysia MADANI, n.d.)*

# Literature Review

## 2.1 Non-Local Means (NLM) Denoising

Non-Local Means (NLM) is an image-denoising technique introduced by Buades et al. (2005) that reduces noise by exploiting similarities between image patches. Unlike conventional local smoothing filters, which estimate a pixel primarily from neighbouring pixels, NLM searches a larger region of the image for patches with similar structural characteristics. Pixels belonging to patches that are more similar to the target patch are assigned greater weights when calculating the restored pixel value. This non-local approach enables noise reduction while preserving important image structures.

NLM is particularly relevant to fingerprint image enhancement because fingerprint images contain repetitive ridge-and-valley patterns. Similar ridge structures can occur at different locations within the fingerprint, allowing NLM to use repeated structural information when estimating pixel intensities. As a result, the technique can suppress image noise while reducing excessive smoothing of ridge boundaries. However, its effectiveness depends strongly on the selected denoising strength. An excessively high filtering strength may remove useful ridge details, whereas an insufficient strength may leave considerable noise in the enhanced image.

In this project, NLM denoising is implemented using OpenCV's Fast Non-Local Means Denoising algorithm on an 8-bit representation of the normalised fingerprint image. The template window and search window are fixed at 7 and 21 pixels respectively, while the denoising-strength parameter,  $h$ , is varied during development. Six candidate values,  $h = 0.03, 0.04, 0.05, 0.06, 0.07$ , and  $0.08$ , are evaluated on the Development Set. Based on the Development Set evaluation,  $h = 0.07$  was selected because it achieved the highest mean PSNR with the lowest mean MSE among the tested configurations, while maintaining a high SSIM value.



## 2.2 Modified Gabor Ridge Enhancement

Gabor filtering is a classical image-processing technique commonly applied to fingerprint enhancement because Gabor filters can respond selectively to specific spatial frequencies and orientations. Fingerprint ridges exhibit locally consistent orientation and frequency characteristics, making orientation-sensitive filtering particularly suitable for strengthening ridge-valley structures. Hong et al. (1998) demonstrated that fingerprint enhancement can be performed by adapting Gabor filters according to locally estimated ridge orientation and frequency, thereby improving ridge clarity in degraded fingerprint regions.

The effectiveness of Gabor filtering depends on the accuracy of the estimated ridge characteristics. When the orientation and frequency estimates correctly represent the local fingerprint structure, the filter can enhance genuine ridge patterns while suppressing components that do not correspond to the expected ridge flow. However, inaccurate orientation or frequency estimation can cause the filter to enhance incorrect structures or modify valid fingerprint details. Gabor-based processing can also be computationally expensive when multiple orientations or parameter combinations must be evaluated.

The Modified Gabor method implemented in this project extends the basic Gabor approach by estimating the fingerprint foreground region, local ridge orientation, coherence, and frequency-related confidence. The Gabor response is then adaptively blended with the fingerprint image according to the estimated ridge information. During development, several parameter combinations involving orientation bins, frequency settings, response strength, sigma scale, and other filtering parameters are evaluated. To reduce computational cost, an initial 50-image subset is used for preliminary screening, followed by evaluation of the best-performing candidates on the full 500-image Development Set. A zero-strength configuration is also included as a diagnostic control to confirm that no enhancement is applied when the response strength is disabled.

### 2.3 Total Variation (TV) Restoration

Total Variation (TV) restoration is an image-denoising approach designed to reduce unwanted intensity variations while preserving significant image boundaries. Unlike conventional linear smoothing techniques, which may blur important edges together with noise, TV restoration controls the overall spatial variation within an image. The original Total Variation model proposed by Rudin et al. (1992) demonstrated its effectiveness for noise removal while maintaining discontinuities, while Chambolle (2004) later introduced an efficient algorithm for solving the associated optimisation problem.

This edge-preserving characteristic makes TV restoration relevant to fingerprint enhancement. Fingerprint images contain narrow and closely spaced ridge-valley structures whose boundaries should remain distinguishable after denoising. Excessive smoothing may merge neighbouring ridges or weaken fine fingerprint details. TV restoration therefore provides a useful balance between suppressing image noise and retaining structurally important boundaries. Nevertheless, its performance is influenced by the regularisation strength. A low regularisation value may provide insufficient denoising, whereas an excessively high value may oversmooth the fingerprint and remove useful ridge information.

In this project, TV restoration is implemented using the Chambolle TV denoising method provided by scikit-image. Four regularisation weights, 0.01, 0.02, 0.04, and 0.06, are evaluated during development. Based on the Development Set results, a weight of 0.02 is selected as the final configuration for subsequent validation. The selected configuration provides a suitable balance between noise reduction and structural preservation, while the project evaluates its performance using PSNR, SSIM, MSE, and runtime.

## 2.4 Coherence-Guided Directional Diffusion

Coherence-Guided Directional Diffusion is based on anisotropic diffusion, in which image smoothing is performed selectively according to local structural information rather than equally in all directions. In contrast to isotropic smoothing, directional diffusion attempts to preserve meaningful directional structures while reducing undesirable variations. Coherence-enhancing diffusion is particularly suitable for images containing flow-like patterns because the direction and strength of local structures can be estimated and incorporated into the diffusion process (Weickert, 1999).

This property is particularly relevant to fingerprint images because fingerprint ridges form locally coherent directional patterns. By estimating local ridge orientation and coherence, smoothing can be directed according to the ridge flow instead of being applied uniformly across ridge boundaries. This can reduce noise while maintaining the directional characteristics of fingerprint ridges. However, the effectiveness of directional diffusion depends on reliable orientation and coherence estimation. Its iterative processing can also increase computational cost compared with simpler denoising techniques.

In this project, local ridge orientation and coherence are estimated before the orientation is quantised into eight directions. Oriented smoothing is then performed for each direction, followed by coherence-weighted iterative updates within the estimated fingerprint foreground. Four combinations of iteration count and step size are evaluated: (2, 0.12), (4, 0.16), (6, 0.14), and (4, 0.22). Based on the Development Set evaluation, 2 iterations with a step size of 0.12 are selected as the final configuration because this setting provides the best overall balance between enhancement quality and processing cost among the tested candidates.

## 2.5 Analysis of the Existing Algorithms (Comparison)

Fingerprint image enhancement can be performed using different classical image-processing approaches, each targeting different types of degradation. Some techniques focus primarily on noise suppression, while others emphasise ridge enhancement, edge preservation, or directional structural restoration. Therefore, the suitability of an enhancement technique depends on its ability to improve image quality while preserving fingerprint ridge information.

Table 2.1 compares the four enhancement techniques selected for this project: Non-Local Means (NLM) Denoising, Modified Gabor Ridge Enhancement, Total Variation (TV) Restoration, and Coherence-Guided Directional Diffusion. The comparison summarises their main advantages, limitations, and relevance to fingerprint enhancement before their quantitative evaluation using PSNR, SSIM, MSE, and processing runtime.

*Table 2.1 Comparison of Selected Fingerprint Enhancement Techniques*

| Algorithm / System               | Example of Previous Work              | Advantages                                                                                                                                                                           | Limitations                                                                                                                                                            | Remarks                                                                                                                                                                                          |
|----------------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Non-Local Means (NLM)            | Buades et al. (2005)                  | Exploits similarities between image patches to reduce noise while retaining repeated structural patterns. Particularly relevant to repetitive fingerprint ridge-and-valley textures. | Performance depends on the denoising-strength parameter. Excessive denoising may remove fine ridge details, while weak denoising may leave residual noise.             | Applied for fingerprint noise reduction. The denoising strength $h$ is tuned while the template and search windows remain fixed. The Development Set selects $h = 0.07$ .                        |
| Modified Gabor Ridge Enhancement | Hong et al. (1998)                    | Uses local orientation and frequency characteristics to enhance ridge-valley structures. Directly targets the directional nature of fingerprint patterns.                            | Sensitive to errors in local orientation and frequency estimation. Evaluation across multiple orientations and parameter combinations can increase computational cost. | Applied to strengthen fingerprint ridge-valley structures using local orientation, coherence, and frequency information. The final configuration is selected through Development Set evaluation. |
| Total Variation (TV) Restoration | Rudin et al. (1992); Chambolle (2004) | Reduces noise while preserving significant image boundaries, helping prevent the excessive edge blurring associated with simple linear smoothing.                                    | The result depends on regularisation strength. Excessive regularisation may oversmooth the image and remove useful ridge information.                                  | Applied for edge-preserving fingerprint restoration using Chambolle TV denoising. The selected regularisation weight is 0.02.                                                                    |

|                                                 |                 |                                                                                                                                                               |                                                                                                                       |                                                                                                                                                                                  |
|-------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coherence-G<br>uidd<br>Directional<br>Diffusion | Weickert (1999) | Performs structure-aware smoothing according to local orientation and coherence, making it suitable for directional and flow-like fingerprint ridge patterns. | Depends on reliable local orientation and coherence estimation. Iterative processing can increase computational cost. | Applied to smooth noise along the estimated fingerprint ridge direction while preserving ridge structure. The selected configuration uses 2 iterations with a step size of 0.12. |
|-------------------------------------------------|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

# Methodology

## 3.1 System flowchart

Figure 3.1 illustrates the experimental workflow used to compare the four fingerprint image enhancement techniques. The process begins with the setup of the core processing and evaluation functions, followed by an audit of the SOCOFing dataset to ensure that the required fingerprint images are available and correctly organised. The dataset is then divided using a deterministic subject-disjoint split into a Development Set containing 500 real fingerprint images from 50 subjects and a Validation Set containing 1,000 real fingerprint images from 100 unseen subjects. This ensures zero subject overlap and zero image overlap between the two sets. This separation ensures that the images used for final validation are independent of those used during parameter selection.

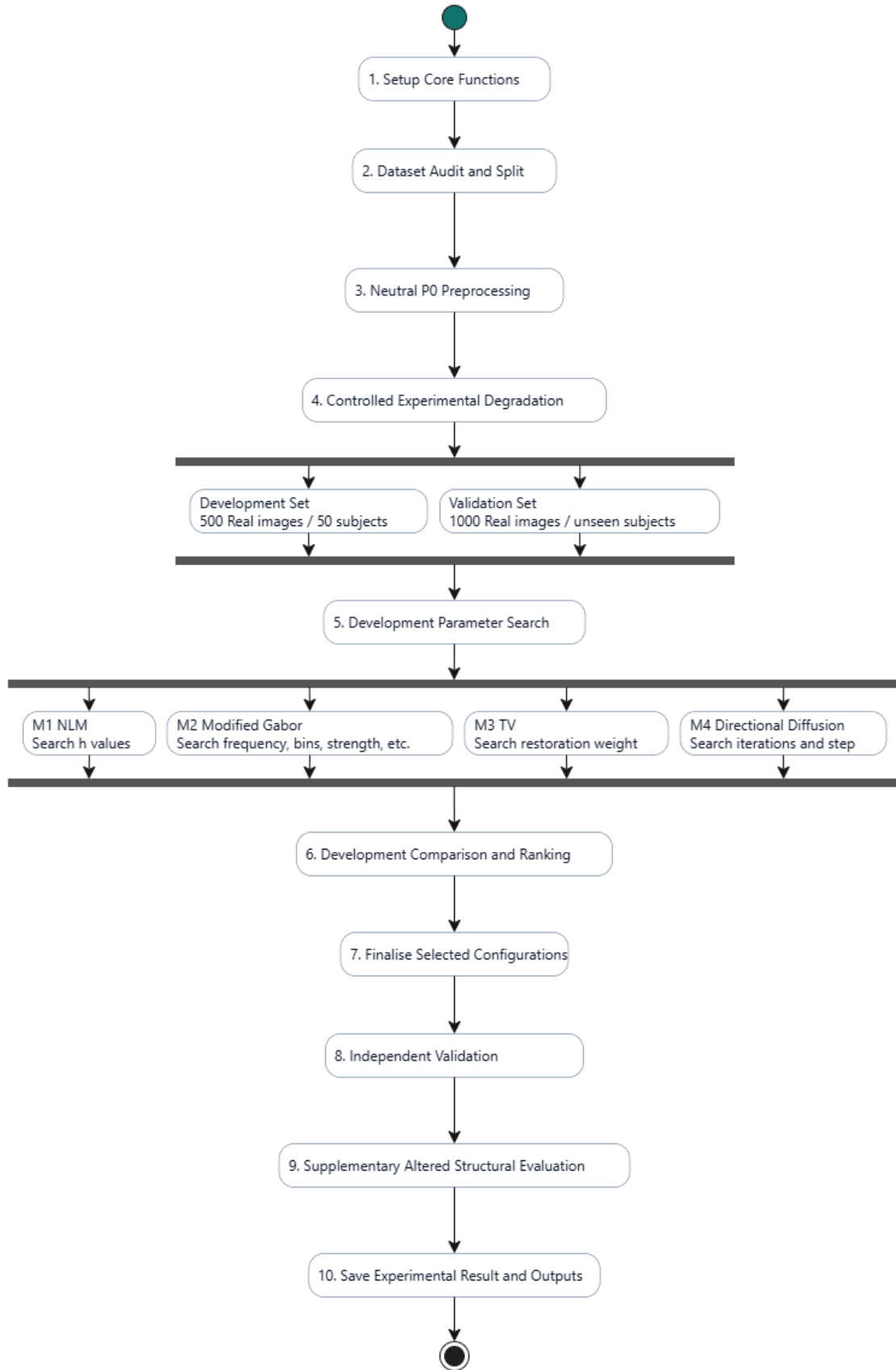
Before applying the enhancement techniques, all selected fingerprint images undergo the same neutral P0 preprocessing, consisting of grayscale conversion, resizing to  $256 \times 256$  pixels, dtype conversion, and intensity normalisation to  $[0,1]$ . No enhancement operation is applied during this stage. A controlled Gaussian-noise degradation is then added to the clean reference images using a fixed severity and deterministic noise seeds. The degraded images are not re-normalised after noise addition, ensuring that the same degraded input is used for all four enhancement techniques.

The Development Set is used exclusively for parameter search and configuration selection. Four enhancement techniques are evaluated independently: Non-Local Means (NLM) denoising, Modified Gabor Ridge Enhancement, Total Variation (TV) Restoration, and Coherence-Guided Directional Diffusion. Different candidate parameter configurations are tested for each technique. NLM searches for different denoising-strength ( $h$ ) values, Modified Gabor evaluates frequency-, orientation- and response-related parameters, TV Restoration evaluates different regularisation weights, and Directional Diffusion evaluates different combinations of iteration count and step size.

The candidate configurations are then compared and ranked according to the defined quantitative evaluation criteria. Based on the Development Set results, the best-performing configuration for each technique is selected and finalised. The overall Development Set winner is also determined at this stage. The Validation Set is then used exclusively for independent evaluation and does not influence parameter selection or change the finalised development decision. The finalised configurations are subsequently applied to the independent Validation Set to evaluate their ability to generalise to fingerprint images from subjects that were not involved in parameter selection.

Following the main validation experiment, the altered fingerprint images available in SOCOFing are used for supplementary qualitative or structural testing. This evaluation is kept separate from the main full-reference experiment because the altered images are not guaranteed to be pixel-aligned with the Real images and are therefore unsuitable for direct PSNR, SSIM, and MSE comparison. Finally, the experimental outputs, selected configurations and evaluation results are saved for subsequent analysis and presentation in Chapter 4.





**Figure 3.1:** *Experimental Workflow for Fingerprint Enhancement Technique Development and Validation.*

### 3.2 Description of dataset

The dataset used in this project is the Sokoto Coventry Fingerprint Dataset (SOCOFing), originally introduced by Shehu et al. (2018). SOCOFing contains 6,000 real fingerprint images from 600 subjects, with ten fingerprint images representing the ten fingers of each subject. The dataset also contains synthetically altered fingerprints organised into three difficulty levels: Altered-Easy, Altered-Medium and Altered-Hard. The alterations include obliteration (Obl), central rotation (CR), and z-cut (Zcut), resulting in 55,273 fingerprint images in total.

The fingerprint images are stored as single-channel 8-bit greyscale BMP files. Each filename contains information describing the fingerprint, including the subject identifier, gender, hand and finger name. Altered-image filenames additionally contain the type of alteration. For example, 100\_\_M\_Left\_index\_finger\_CR.BMP represents the left index fingerprint of male Subject 100 with a central rotation alteration.

For the main quantitative experiment, the Real images are divided at the subject level into two independent subsets. The Development Set consists of 500 fingerprint images from 50 subjects, while the Validation Set consists of 1,000 fingerprint images from 100 unseen subjects. Each selected subject contributes ten fingerprint images. The Development Set is used for parameter searching and configuration selection, whereas the Validation Set is reserved for evaluating the selected configurations. There is no subject or image overlap between the two sets, ensuring that the validation images remain independent from the parameter-selection process.

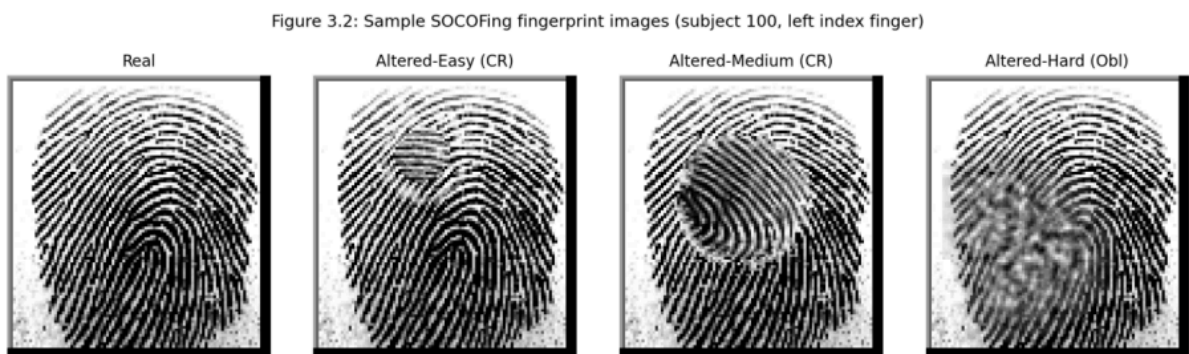
Before controlled degradation, every Real image undergoes the same neutral P0 preprocessing. The image is converted to greyscale, resized to  $256 \times 256$  pixels, converted to a floating-point representation, and intensity-normalised to the range  $[0,1]$ . No enhancement, restoration, or member-specific processing is applied during this stage. This ensures that no member technique receives an advantage from the shared preprocessing stage.

For paired full-reference evaluation, a controlled degradation is generated from each clean P0 fingerprint by adding reproducible Gaussian noise. A fixed experimental severity of 0.55 is used, corresponding to a Gaussian noise standard deviation of approximately 0.0483 in the normalised  $[0,1]$  intensity range. Deterministic random seeds are used to make the degradation reproducible, and the degraded image is not renormalised after noise addition. Most importantly, the same degraded fingerprint is supplied to all four enhancement techniques, allowing their outputs to be compared under identical input conditions.

The clean P0 image is retained as the reference for each degraded image. After enhancement, the output can therefore be directly compared with its corresponding clean reference using Mean Squared Error (MSE), Peak Signal-to-Noise Ratio (PSNR), and Structural Similarity Index (SSIM). This paired design enables quantitative measurement of how closely each enhancement technique restores the degraded fingerprint towards its clean reference.

The Altered-Easy, Altered-Medium and Altered-Hard subsets are used separately for supplementary structural evaluation. Since these altered fingerprints are not treated as pixel-aligned degraded versions of the Real reference images, they are not used for the main full-reference PSNR, SSIM and MSE comparison. This separation prevents unsupported full-reference measurements from being reported for unpaired altered fingerprints.

Dataset source: Sokoto Coventry Fingerprint Dataset (SOCOFing), available at <https://www.kaggle.com/datasets/ruizgara/socofing> and originally described by Shehu, Ruiz-Garcia, Palade and James (2018) (arXiv:1807.10609). Per the assignment specification, the dataset is cited rather than bundled with the submitted archive.



**Figure 3.2:** Sample SOCOFing fingerprint images (subject 100, left index finger).

### 3.3 Applications of the algorithm(s)

The four fingerprint enhancement techniques are implemented as independent Python image-processing functions within a common experimental pipeline. Each technique receives the same neutral-preprocessed and degraded fingerprint input to ensure a fair comparison. The pipeline performs fingerprint preprocessing, controlled degradation, enhancement, quantitative evaluation, and output generation.

The Non-Local Means (NLM) Denoising technique is implemented using OpenCV's `fastNlMeansDenoising()` function. The degraded fingerprint is converted to an 8-bit image before denoising. The template and search window sizes are fixed at  $7 \times 7$  and  $21 \times 21$  pixels respectively, while the denoising-strength parameter  $h$  is varied during development. Six candidate values from 0.03 to 0.08 are evaluated before the selected configuration is applied to validation images.

The Modified Gabor Ridge Enhancement technique enhances fingerprint ridge-valley structures using estimated foreground regions, local ridge orientation, coherence, and frequency-related information. Multiple parameter configurations are evaluated, including different frequency settings, orientation bins, response strengths, and filter parameters. Due to the larger parameter space, candidate configurations are first screened on a 50-image subset before the best candidates are evaluated on the full Development Set.

The Total Variation (TV) Restoration technique is implemented using the Chambolle TV denoising algorithm from `scikit-image`. Four regularisation weights, 0.01, 0.02, 0.04, and 0.06, are evaluated to determine a suitable balance between noise reduction and preservation of fingerprint ridge boundaries.

The Coherence-Guided Directional Diffusion technique performs iterative smoothing according to the estimated fingerprint ridge direction and coherence. The local orientation field is quantised into eight directions, and coherence-weighted smoothing is applied within the fingerprint region. Four combinations of iteration count and step size are evaluated during development.

For each candidate configuration, the enhanced fingerprint is compared with its corresponding clean reference using PSNR, SSIM, MSE, and processing runtime. The selected configuration for each technique is then finalised and applied to the independent Validation Set. The final configurations are also applied to samples from the SOCOFing altered subsets for supplementary structural evaluation.

**Algorithm 1:** Comparative Fingerprint Enhancement Experiment

Input:

SOCOFing Real fingerprint images  
SOCOFing Altered fingerprint images

Output:

Selected configuration for each enhancement technique  
Development and validation evaluation results  
Enhanced fingerprint images

1. Audit SOCOFing dataset
2. Split Real fingerprint images into subject-disjoint Development and Validation Sets.
3. FOR each selected Real fingerprint:
  - clean  $\leftarrow$  P0\_Preprocess(image)
  - degraded  $\leftarrow$  Add\_Controlled\_Gaussian\_Noise(clean)
  - Store clean and degraded pair
 END FOR
4. FOR each enhancement technique:
  - IF technique = NLM:
    - Search  $h \in \{0.03, 0.04, 0.05, 0.06, 0.07, 0.08\}$
  - ELSE IF technique = Modified Gabor:
    - Screen candidate configurations on 50 development images
    - Select top 3 non-identity candidates

```

    Evaluate top 3 on all 500 development images
ELSE IF technique = TV Restoration:
    Search weight  $\in \{0.01, 0.02, 0.04, 0.06\}$ 
ELSE IF technique = Directional Diffusion:
    Search (iterations, step)  $\in$ 
         $\{(2,0.12), (4,0.16), (6,0.14), (4,0.22)\}$ 
FOR each candidate configuration:
    Apply technique to every Development degraded image
    Calculate PSNR, SSIM, MSE and runtime
END FOR

```

Select best configuration using:

1. Highest mean PSNR
2. Highest mean SSIM
3. Lowest mean MSE

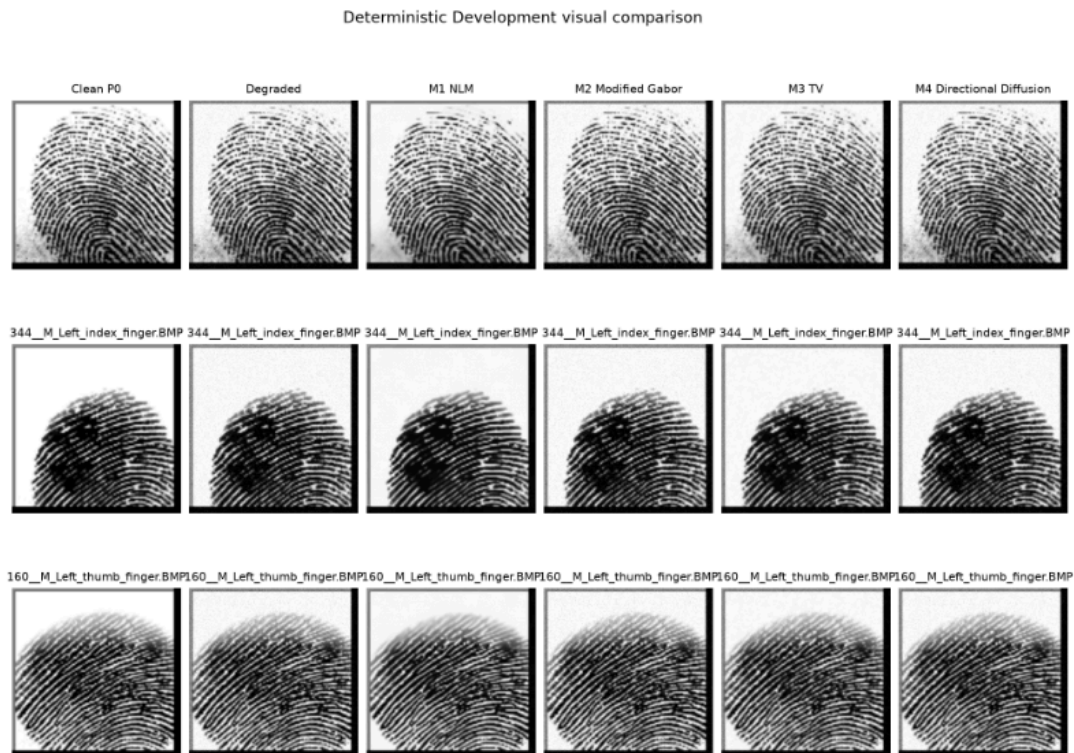
END FOR

5. Finalise the selected configuration for each technique before validation.
6. Apply all finalised configurations to the independent Validation Set
7. Calculate validation PSNR, SSIM, MSE and runtime
8. Apply finalised configurations to sampled  
Altered-Easy, Altered-Medium and Altered-Hard images
9. Calculate supplementary structural measurements
10. Save experimental metrics, configurations,  
figures and enhanced outputs

## Sample Input and Output

**Sample Input:** A SOCOFing fingerprint image after P0 preprocessing and controlled Gaussian-noise degradation.

**Sample Output:** Enhanced fingerprint images generated by NLM Denoising, Modified Gabor Ridge Enhancement, TV Restoration, and Coherence-Guided Directional Diffusion.



*Figure 3.3: Sample input and output comparison for the four fingerprint enhancement techniques.*

# Result & Discussion

## 4.1 Experimental Results

The four fingerprint enhancement techniques were evaluated using the experimental procedure described in Chapter 3. Each technique was first tuned on the 500-image Development Set, and the selected configuration was then finalised before independent evaluation on the 1,000-image Validation Set. A degraded baseline was included to measure the improvement produced by each enhancement technique. The main quantitative evaluation used Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM), Mean Squared Error (MSE), and processing runtime.

Table 4.1 summarises the selected configuration for each enhancement technique after Development Set parameter searching. NLM selected a denoising strength of  $h = 0.07$ , TV Restoration selected a regularisation weight of 0.02, and Coherence-Guided Directional Diffusion selected 2 iterations with a step size of 0.12. The Modified Gabor configuration was selected through its two-stage Development Set parameter search. All selected configurations were finalised before validation and were not modified using Validation Set results.

Table 4.2 presents the quantitative comparison on the 500-image Development Set. NLM achieved the highest mean PSNR of 29.568 dB and the lowest mean MSE of approximately 0.001121 among the four enhancement techniques. TV Restoration achieved the highest mean SSIM of 0.9379 and the lowest mean processing time. Directional Diffusion produced a moderate improvement over the degraded baseline, whereas Modified Gabor produced only a small improvement under the controlled Gaussian-noise degradation used in the experiment.

Figure 4.1 provides a qualitative comparison of selected Development Set samples before and after enhancement. The figure shows the clean P0 reference, degraded input, and the corresponding outputs produced by NLM, Modified Gabor, TV Restoration, and Coherence-Guided Directional Diffusion.



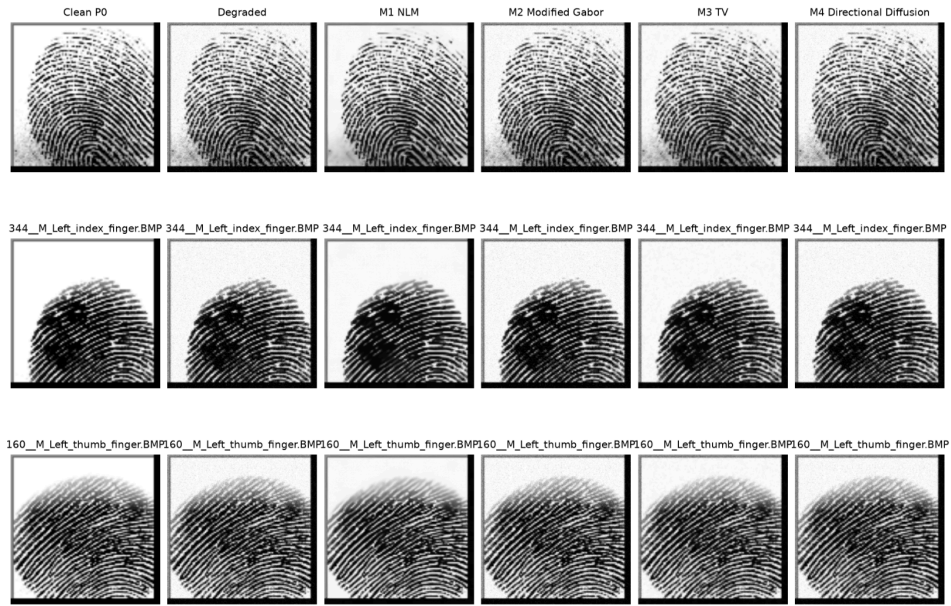
*Table 4.1: Development-Selected Configuration for Each Enhancement Technique*

| Member Technique      | Development-Selected Configuration                          | Selection Status |
|-----------------------|-------------------------------------------------------------|------------------|
| NLM                   | $h = 0.07$                                                  | Selected         |
| Modified Gabor        | Selected through two-stage Development Set parameter search | Selected         |
| TV                    | Weight = 0.02                                               | Selected         |
| Directional Diffusion | Iterations = 2; step = 0.12                                 | Selected         |

*Table 4.2: Development Set Performance Comparison of the Four Enhancement Techniques*

| Method                   | Images | Mean PSNR (dB) | Mean SSIM | Mean MSE | Delta PSNR | Delta SSIM | Mean Runtime |
|--------------------------|--------|----------------|-----------|----------|------------|------------|--------------|
| Degraded Baseline        | 500    | 27.357         | 0.8415    | 0.001855 | -          | -          | -            |
| M1 NLM                   | 500    | 29.568         | 0.9336    | 0.001121 | 2.2110     | 0.0921     | 0.480        |
| M2 Modified Gabor        | 500    | 27.367         | 0.8419    | 0.001850 | 0.0102     | 0.0004     | 0.754        |
| M3 TV                    | 500    | 29.287         | 0.9379    | 0.001203 | 1.9303     | 0.0964     | 0.252        |
| M4 Directional Diffusion | 500    | 27.986         | 0.8511    | 0.001601 | 0.6296     | 0.0096     | 0.774        |

# Deterministic Development visual comparison



*Figure 4.1: Development Set visual comparison of the degraded fingerprint input and outputs from the four enhancement techniques.*

## 4.2 Discussion and Critical Interpretation

Based on the predefined Development Set ranking rule, Non-Local Means (NLM) is selected as the overall best-performing technique in this project. The ranking prioritises the highest mean PSNR, followed by SSIM and MSE. NLM achieved the highest Development Set mean PSNR of 29.568 dB and was therefore selected and finalised before validation. Its performance remained stable on the independent Validation Set, where it achieved a mean PSNR of 29.559 dB. This indicates that the selected NLM configuration generalised well to fingerprint images from unseen subjects rather than only performing well on the Development Set.

Although NLM is the overall project winner, Total Variation (TV) Restoration provides the strongest alternative when structural similarity and processing efficiency are prioritised. TV achieved the highest mean SSIM on both the Development and Validation Sets and also recorded the lowest processing runtime among the four enhancement techniques. On the Validation Set, TV achieved a mean SSIM of 0.9358 compared with 0.9300 for NLM, while requiring approximately 0.254 seconds per image. Therefore, the results demonstrate that the definition of “best” depends partly on the evaluation priority: NLM performs best according to the project’s predefined PSNR-first ranking, while TV performs better in terms of SSIM and runtime.

The strong performance of NLM can be explained by the controlled degradation used in the experiment. The paired Real-image experiment introduces additive Gaussian noise, while preserving the underlying ridge geometry. NLM is specifically designed to suppress noise by exploiting similarities between image patches, and fingerprint images contain repeated ridge-and-valley patterns that provide suitable patch structures. As a result, NLM is well matched to the degradation model used in the experiment.

The Modified Gabor Ridge Enhancement technique produced only a small quantitative improvement over the degraded baseline. However, this result should not be interpreted as evidence that Gabor filtering is unsuitable for fingerprint enhancement. The controlled degradation mainly introduces Gaussian noise rather than severe ridge-orientation or frequency distortion. Since Modified Gabor processing is designed to strengthen ridge structures using local orientation, coherence, and frequency information, the experiment

provides limited opportunity for this structure-specific processing to improve full-reference PSNR without also modifying valid ridge intensities.

Coherence-Guided Directional Diffusion produced a moderate improvement over the degraded baseline but required the highest processing time among the four methods. On the Validation Set, it improved mean PSNR from 27.363 dB to 27.992 dB but required approximately 0.772 seconds per image. This suggests that its structure-aware iterative processing introduces additional computational cost, while providing less benefit under the Gaussian-noise degradation used in the current experiment.

The supplementary Altered-Easy, Altered-Medium, and Altered-Hard evaluation is kept separate from the main paired experiment because these altered fingerprint images do not provide suitable pixel-aligned clean references. Therefore, PSNR, SSIM, and MSE are not used to claim restoration accuracy on these subsets. Instead, the altered-image experiment provides supplementary structural evidence only. This prevents unsupported full-reference conclusions from being drawn from unpaired fingerprint images.

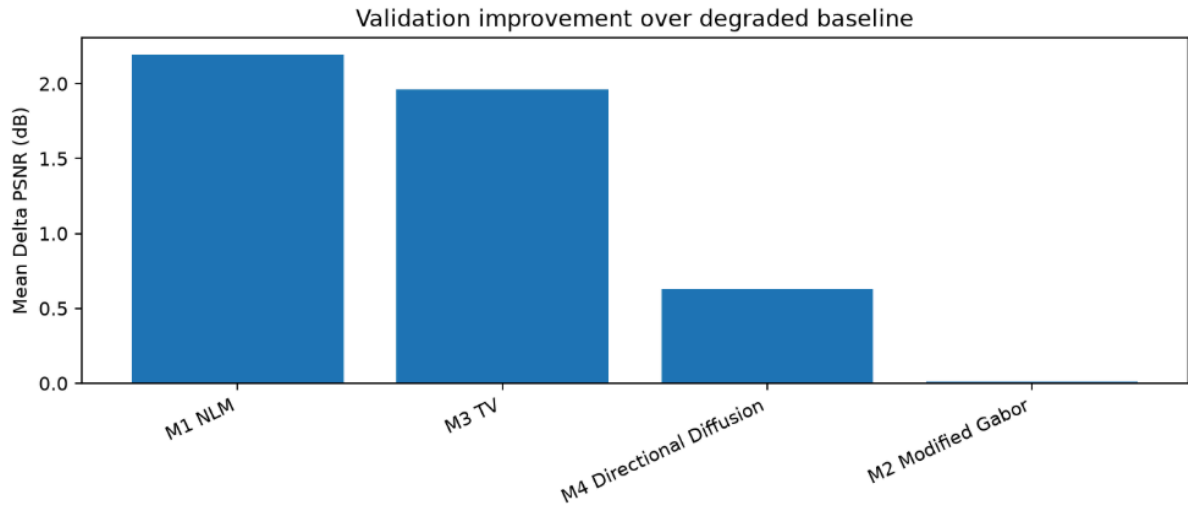


Figure 4.2: Validation Set PSNR improvement of the four enhancement techniques over the degraded baseline.

# Conclusion

## 5.1 Achievements

This project successfully implemented and evaluated four classical fingerprint image enhancement techniques: Non-Local Means (NLM) Denoising, Modified Gabor Ridge Enhancement, Total Variation (TV) Restoration, and Coherence-Guided Directional Diffusion. A controlled experimental framework was developed using the SOCOFing dataset, with a subject-disjoint Development Set of 500 fingerprint images from 50 subjects and an independent Validation Set of 1,000 fingerprint images from 100 unseen subjects. This design ensured that parameter selection and final validation were conducted using separate subjects and images.

The experimental results demonstrated that the four techniques provide different advantages depending on the evaluation criterion. Based on the predefined PSNR-first ranking, NLM was identified as the overall best-performing technique. With the selected configuration of  $h = 0.07$ , NLM achieved a mean PSNR of 29.568 dB on the Development Set and maintained a similar mean PSNR of 29.559 dB on the independent Validation Set. This consistent performance indicates that the selected NLM configuration generalised effectively to fingerprint images from unseen subjects.

TV Restoration also demonstrated strong performance and provided an important alternative to NLM. Although its PSNR was slightly lower, TV achieved the highest mean SSIM on both the Development and Validation Sets and recorded the lowest processing runtime among the four techniques. Therefore, the results demonstrate that the most suitable enhancement technique depends on the intended priority: NLM is preferable when pixel-level reconstruction fidelity measured by PSNR is prioritised, whereas TV Restoration is attractive when structural similarity and processing efficiency are more important.

Coherence-Guided Directional Diffusion produced a moderate improvement over the degraded baseline, while Modified Gabor Ridge Enhancement produced only a small quantitative improvement under the controlled Gaussian-noise degradation used in this study. These results should be interpreted in relation to the experimental degradation model rather than as evidence that structure-aware techniques are generally unsuitable for fingerprint enhancement. In particular, the experiment primarily introduced Gaussian noise while preserving the underlying ridge geometry, which is well matched to the denoising characteristics of NLM.

Overall, the project achieved its main goal of providing a controlled and reproducible comparison of classical fingerprint enhancement techniques. The final system also provides a practical Streamlit

prototype through which the selected enhancement methods can be applied to fingerprint images, supporting the practical use of the experimental findings.

### **Objective Achievement**

Objective 1 status: Fulfilled: All four fingerprint enhancement techniques were successfully implemented and evaluated using the 500-image Development Set. Their performance was compared using PSNR, SSIM, MSE and processing runtime, and suitable configurations were identified for each technique.

Objective 2 status: Fulfilled: The configuration of each technique was selected and finalised based only on Development Set results before being evaluated on the independent 1,000-image Validation Set from 100 unseen subjects. No subject or image overlap occurred between the Development and Validation Sets, and the validation results did not influence parameter selection.

Objective 3 status: Fulfilled: A practical Streamlit prototype was developed to apply the selected fingerprint enhancement techniques to uploaded images. The prototype supports single-image processing, comparison of all four techniques and bulk-processing workflows, while reference-based metrics are reported only when an appropriate reference image is provided.

**Project Code Repository:** <https://github.com/mol2k/ImageProcessing.git>  
<https://fingerprint-enhancement-jupyter-projectfinge-cz2xz3.streamlit.app/>

## 5.2 Limitations and Future Works

Although the project achieved its stated objectives, several limitations should be considered when interpreting the experimental results. First, the main quantitative experiment uses a controlled Gaussian-noise degradation to create paired degraded and clean fingerprint images. This approach enables reliable calculation of full-reference metrics such as PSNR, SSIM and MSE; however, Gaussian noise alone does not represent all degradation encountered in practical fingerprint acquisition. Real fingerprint images may contain blur, uneven illumination, smudging, partial ridge loss, sensor artefacts, dry or wet skin effects and other complex distortions. Therefore, the relative performance observed in this experiment may change under different degradation conditions.

Second, the primary quantitative comparison relies on PSNR, SSIM and MSE. These metrics are useful for measuring similarity between an enhanced image and its clean reference but do not directly measure biometric recognition performance. An image with improved PSNR or SSIM does not necessarily produce better fingerprint matching or minutiae extraction. The supplementary Altered-Easy, Altered-Medium and Altered-Hard evaluation provides additional structural evidence, but these altered images are not treated as pixel-aligned reference pairs and therefore cannot be evaluated using the same full-reference metrics.

Third, the parameter search investigates a predefined range of configurations for each enhancement technique rather than every possible parameter combination. More extensive optimisation could potentially identify configurations that achieve better results. Modified Gabor Ridge Enhancement in particular has a comparatively large parameter space involving orientation, frequency and filter-response settings, making exhaustive searching computationally expensive.

Future work could extend the experiment by evaluating the four enhancement techniques under multiple controlled degradation types and severity levels, including Gaussian noise, blur, contrast reduction and combinations of these degradations. This would provide a broader assessment of the conditions under which each technique performs best. Further evaluation could also incorporate fingerprint-specific measures such as ridge continuity,

minutiae preservation or actual fingerprint matching accuracy to complement PSNR, SSIM and MSE.

The computational performance of the structure-aware techniques could also be improved through algorithm optimisation, parallel processing or GPU acceleration. In addition, future work could compare the classical techniques investigated in this project with modern learning-based fingerprint enhancement approaches. Such extensions would provide a more comprehensive understanding of the trade-offs among image quality, ridge preservation, computational efficiency and biometric recognition performance.



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# Appendix A

## TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY Group Contract Form

Faculty : Faculty of Computing and Information Technology (FOCS)  
Programme : RSW  
Year of Study : 2  
Semester : 3  
Course Title : Image Processing  
Course Code : BMDS2133  
Group Assignment / Project Title : Fingerprint Image Enhancement and Quality Assessment System  
Name of Tutor : Nurul Akmar Binti Azman  
Tutorial Group : 1

| Student Name and ID          | Role(s)/Task(s) Assigned                  | Signature and Date                                                                                  |
|------------------------------|-------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Mok Chun Bing<br>24WMR13492  | Non-Local Means (NLM)<br>Denoising        | <br>20/08/2026   |
| Lee Mun Hinn<br>24WMR16749   | Modified Gabor Ridge<br>Enhancement       | <br>20/08/2026 |
| Lam Zhan Foong<br>24WMR13805 | Total Variation (TV)<br>Restoration       | <br>20/08/2026 |
| Lim Kai Xi<br>24WMR12665     | Coherence-Guided Directional<br>Diffusion | <br>20/08/2026 |

# Appendix B

## AI Usage Disclosure Form

**Student Name** : Lee Mun Hinn

**Programme** : RSW

**Course** : BMDS2133 Image Processing

**Assessment Title** : Fingerprint Image Enhancement and Quality Assessment System

### 1. AI Tool(s) Used (Check that all that apply)

- ☐ None (I did not use any AI tools for this assignment)
- ☒ ChatGPT (Version: 5.6 Sol)
- ☐ Deepseek
- ☐ Gemini
- ☐ Other: \_\_\_\_\_

### 2. Nature of Assistance (Check all that apply)

- ☒ **Brainstorming:** Generating ideas, topics, or outlines.
- ☒ **Research:** Summarising long articles or finding concepts.
- ☒ **Editing:** Checking grammar, spelling, or sentence structure.
- ☐ **Coding/Math:** Debugging code or explaining a formula.
- ☐ **Creation:** Generating images, data, or draft text.

### 3. Disclosure Statement

*AI tools (ChatGPT) were used to assist with brainstorming project ideas, improving grammar and sentence structure, summarising background information, and refining the proposal. All project concepts, system design, analysis, and final decisions were reviewed, verified, and modified by the group members to ensure that the submitted work reflects our own understanding and independent effort.*

### 4. Verification & Integrity

- ☒ I have reviewed and verified the accuracy of all facts, data, and citations generated with the assistance of AI tools.
- ☒ I have properly acknowledged and cited any AI-generated text, ideas, or content in accordance with my instructor's requirements.
- ☒ I confirm that the reasoning, interpretation, and analysis presented in the final submission are entirely my own.

**Student Signature** : Lee

**Date** : 18/08/2026